

2.4 - Product and Quotient Rule

Goals: Compute derivatives using the product and quotient Rules, including of trigonometric functions

Before Class Videos: 2.4.1 and 2.4.2

Theorem (Derivative Rules):

- $\frac{d}{dx}(c) = 0$ where c is a constant.
- $\frac{d}{dx}(x^n) = nx^{n-1}$ where $n > 0$ is an integer.

If $f(x)$ and $g(x)$ are differentiable at $x = a$, then

- $\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x)$
- $\frac{d}{dx}[cf(x)] = c\frac{d}{dx}f(x)$

Let's add two new rules to the list

Theorem (Product and Quotient Rules):

$$\frac{d}{dx}[fg] = f'g + fg'$$
$$\frac{d}{dx}\left[\frac{f}{g}\right] = \frac{f'g - fg'}{g^2}$$

Example 1: Compute the following derivatives

a)

$$\sqrt{x}e^x$$

b)

$$\frac{\sin(x)}{x^3}$$

c)

$$\frac{xe^x}{x^2 + 1}$$

Example 2: Let's explore the intuition behind why the product rule is NOT just the product of the derivatives.

Example 3: Let's finally figure out what exactly e^x is all about. We'll use the definition of the derivative to compute, for $a > 0$,

$$\frac{d}{dx}a^x$$

Let's explore

$$\lim_{h \rightarrow 0} \frac{a^h - 1}{h}$$

Definition:

The number e is defined to be the value of a such that $\lim_{h \rightarrow 0} \frac{a^h - 1}{h}$

Right-Triangle Definitions

$$\begin{array}{lll} \sin x = \frac{\text{opposite}}{\text{hypotenuse}} & \tan x = \frac{\text{opposite}}{\text{adjacent}} & \sec x = \frac{\text{hypotenuse}}{\text{adjacent}} \\ \cos x = \frac{\text{adjacent}}{\text{hypotenuse}} & \cot x = \frac{\text{adjacent}}{\text{opposite}} & \csc x = \frac{\text{hypotenuse}}{\text{opposite}} \end{array}$$

Rewriting with Sine and Cosine

$$\begin{array}{lll} \sin x = \sin x & \tan x = \frac{\sin x}{\cos x} & \sec x = \frac{1}{\cos x} \\ \cos x = \cos x & \cot x = \frac{\cos x}{\sin x} & \csc x = \frac{1}{\sin x} \end{array}$$

Example 4: Let's explore the graphs of these functions.

Example 5: Compute $\frac{d}{dx} \tan(x)$

Example 6: Compute $\frac{d}{dx} \sec(x)$

Trigonometric Derivatives

$$\begin{array}{lll} \frac{d}{dx} \sin x = \cos x & \frac{d}{dx} \tan x = \sec^2 x & \frac{d}{dx} \sec x = \sec x \tan x \\ \frac{d}{dx} \cos x = -\sin x & \frac{d}{dx} \cot x = -\csc^2 x & \frac{d}{dx} \csc x = -\csc x \cot x \end{array}$$